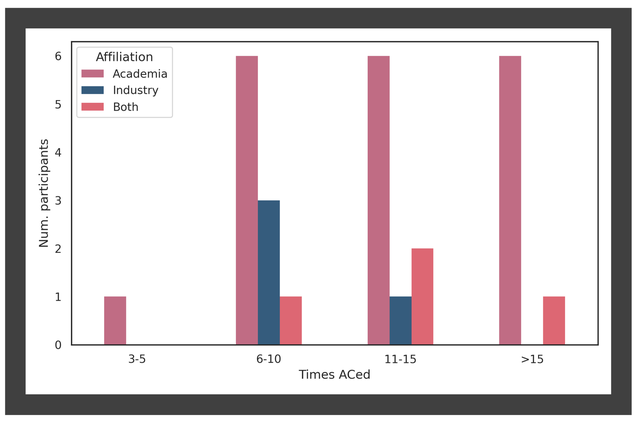
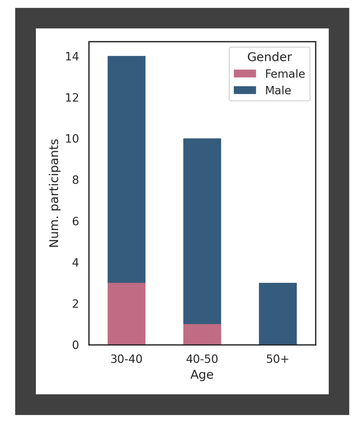




semi-structured interviews with area chairs in ML

n = 27
conducted 2023



in CS conferences
"area chairs"
=
editors
in the journal system

AI support in

peer review

understanding *area chair practices* and *design implications* for AI tools

main findings

neutrality is important for some; other area chairs want to get *involved*

You're sitting there as **kind of a witness** between a discussion between between the authors and the reviewers.

Just **like a judge** is not supposed to enter the arena,… and argue for or against one case, and needs to remain impartial.

I think **my responsibility is towards the paper** and not to either of the parties, and if the idea is is good and it's presented right, then… I try and fight for it.

There are times when I think all the reviewers are wrong, and I disagree with them. **Is it still my job** to, like, maybe try to **overturn** the reviewers?

Design implications: Tools must be **lightweight**, and compatible with diverse area chair workflows. However, **explainability**, and **fidelity to the underlying text**, are seen as crucial by both hands-on and hands-off ACs.

logistical and *social* tasks feel the most burdensome

[Ideally, I would want] a robot that could **hunt down my reviewers** and beat them or a way more **annoying alert system** to get people to do the **work they're supposed to do**.

It's sort of **running back and forth digitally between folks**, and I think it'd be much nicer if reviewers engaged more without the AC having to mediate.

You don't need actually a lot of subject-specific knowledge, and it is a bit of a **waste of time**.

Design implications: An opportunity for **conversational agents**: acting as moderators, agents could handle routine communications such as **encouraging participation** and **following up** on late reviews, or even just **drafting contextually appropriate messages** for these tasks.

generalizing to other fields and venues

- *Overload*: **increasing submissions** and **scarce reviewers** are widely reported.
  → Findings about **logistical** burdens are **likely to generalize** across fields.
- *OpenReview*: ICLR encourages **extended**, **public** peer review discussions.
  → Some findings **may not apply** in conventional peer review settings with **fewer, more private interactions**.
- *Experience*: **ACs in ML** are more well-informed about **potential failure modes** of AI tools.
  → Their cautious stance can be read as a **safety-first baseline** for AI adoption in peer review.

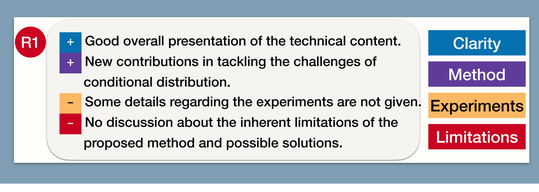
design probe

Four *wireframes* of AI-powered tools to assist area chairs.

Broadly, participants were **cautious** and emphasized the importance of **human oversight**, suggesting that human-in-the-loop tools are apt.

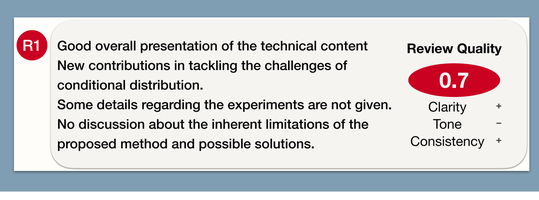
Design implications: Tools should be **flexible** and **configurable** to accommodate different levels of comfort with AI involvement.

Sentence-level sentiment and topic



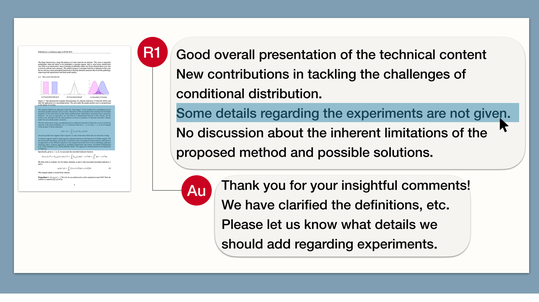
I would prefer that any automated judgment come as early as possible in the process, so that there's as many [human] eyeballs as possible.

Review quality ratings



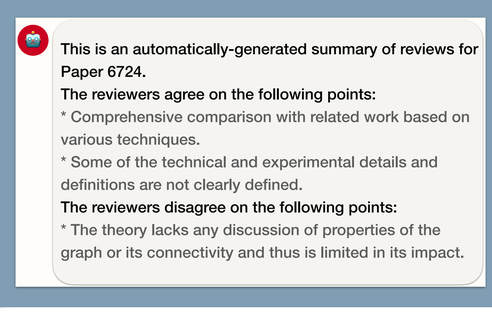
Especially as somebody who does [ML], I wanna know where these numbers came from.

Link review comment → paper context



It could resolve some ambiguity. But I think I think I could imagine extra problems that could introduce.

LLM-generated summary of reviews



That would simplify my life a lot.

If it's automatically generated, will I get too lazy? And if I get too lazy, then am I actually doing my job as an AC?

Hands-Off or Hands-On? Variation in Area Chair Practices and Implications for AI Support

Ines Arous[1], Neha Nayak Kennard[2], Andrei Mircea[3,5], Emily Kuang[1], Jackie Chi Kit Cheung[4,5], Andrew McCallum[2]

[1]York University [2]University of Massachusetts Amherst [3]University of Montreal [4]McGill University, Canada CIFAR AI Chair [5]Mila